

Rudra Karmakar

Frontend Engineer | Python & AI

Kolkata, India | +91 8597442774 | rudrakarmakar666@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Frontend Engineer with 2+ years of experience building responsive, production-grade web applications with **React.js**, **JavaScript (ES6+)**, and **REST APIs**. Skilled in reusable component architecture, form validation, and cross-team collaboration in Agile environments. Hands-on experience with **Microsoft Azure** cloud deployment and CI/CD via **Azure DevOps**, plus work spanning **Next.js**, **TypeScript**, **Python/FastAPI**, secure authentication (**JWT**, **RBAC**), and applied **AI / LLM integration** (OpenAI SDK, NVIDIA NIM, RAG-style chatbots). Microsoft Certified: Azure Fundamentals (AZ-900).

Experience

Tata Consultancy Services (TCS)

Aug 2024 – Present

System Engineer – Frontend Engineer (React.js)

Kolkata, India

- Developed and maintained responsive, reusable, scalable React.js components and REST API integrations for enterprise web applications, including EasyQuote and MyAccount, used for customer gas and electricity plan management.
- Built pixel-accurate, responsive layouts directly from Figma mockups using hand-written CSS/HTML5 alongside **Tailwind CSS** utility classes, ensuring consistent rendering across breakpoints and browsers.
- Used **Cursor** and **GitHub Copilot** as a default part of the daily workflow to scaffold components and API integration boilerplate faster, then reviewed, debugged, and overrode generated state-management and validation logic that didn't match edge cases.
- Collaborated with senior developers and cross-functional stakeholders in an Agile/Scrum environment to plan, build, and ship new features on schedule.

Projects

EasyQuote Web Application | *React.js, TypeScript, Tailwind CSS, REST APIs, Azure*

Oct 2025

- Maintained and enhanced the EasyQuote platform for customer gas/electricity plan management, designing and developing reusable, scalable React.js components across fuel-connection and quotation workflows end to end.
- Optimized form validation logic and shipped 4+ new features, cutting manual steps across 2+ request types and improving processing throughput.
- Implemented React Router-based navigation and url restriction based on authorization and REST API integrations to support evolving business requirements and ensure responsive, user-friendly interfaces.

MyAccount Web Application | *React.js, Redux Toolkit, REST APIs*

Apr 2025

- Built reusable React.js components with Redux Toolkit for state management, validation, and responsive design for customer account, billing, and service-preference management.
- Integrated REST APIs with backend teams to streamline data handling across account-management screens and resolved UI/usability bugs to improve customer experience.

Nemotron AI Chat | *React 19, Vite, FastAPI, NVIDIA NIM, LLM/RAG, OpenAI SDK*

Aug 2026

- Independently designed and built a full-stack AI chat application end to end – a React 19 + Vite frontend and a Python/FastAPI backend – as a self-directed project to learn applied AI engineering.
- Integrated the **OpenAI SDK** against **NVIDIA NIM's** OpenAI-compatible API to serve the **NVIDIA Nemotron LLM**, handling streaming responses and environment-based secret management for API keys.
- Implemented **RAG** (retrieval-augmented generation) using **PostgreSQL** with the **pgvector** extension as the vector store, embedding and retrieving context to ground chatbot responses.
- Persisted chat history in a normalized **conversations/messages** schema and managed frontend state with **Redux**, enabling per-turn context continuity within LLM token limits alongside responsive UI/UX (loading/error handling, message state) and the API contract from scratch.

Customer Insights & Reporting Platform (CIRMP) | *Next.js, TypeScript, FastAPI, JWT, Redis, LLM/RAG*

Aug 2026

- Built the **Next.js/TypeScript** frontend and a supporting **Python/FastAPI** service for an enterprise reporting platform covering customer data entry, validation, Excel/PDF report generation, and a configurable review-approval workflow.
- Backed the platform with a **Pydantic**-validated REST API secured via **JWT/RBAC**, a **Redis** job queue for report generation, and proper **CORS** configuration.
- Embedded my self-built **Nemotron AI Chat** service into the platform as a Generative AI assistant, extending it to ground responses in application data via **RAG** and surface automated, context-aware summaries directly inside generated reports.
- Built reusable UI components handling API state, loading/error states, and end-to-end form validation.

Education

Jalpaiguri Government Engineering College

June 2024

B.Tech, Information Technology – CGPA: 8.5/10

Jalpaiguri, West Bengal

Technical Skills

Frontend: React.js, Next.js, SolidJS, ShadCN, TypeScript, JavaScript (ES6+), React Router, Redux Toolkit, React Context API, Responsive Web Design, Cross-Browser Compatibility

AI / ML: OpenAI SDK, NVIDIA NIM, NVIDIA Nemotron, LLM Integration, AI Chatbots, RAG (Retrieval-Augmented Generation), Vector Databases

Python & APIs: Python, FastAPI, REST APIs, Pydantic, WebSocket, Socket.IO, JWT Authentication

Languages: JavaScript (ES6+), TypeScript, Python, HTML5, CSS3, SQL (PostgreSQL)

Cloud & Tools: Microsoft Azure, Docker, Git, GitHub, GitLab, CI/CD, VS Code, Cursor, Postman, Figma, Swagger

Practices: Agile/Scrum, Unit Testing, OOP & SOLID Principles, AI Integration, Technical Writing & Presentation

Certifications & Achievements

AZ-900: Microsoft Azure Fundamentals – Microsoft

Secured **2nd Position** at **Hack4Bengal** hackathon for building a real-time chat application

Organizer, **JWOC-23** open-source event – 1,500+ participants

Coding profiles: [GitHub](#) | [LeetCode](#)